Random sequential packing of sinusoidal worldlines on a closed conformal loop: jam-count exponents, emergent homogeneity, and the phase-blindness of the packed state

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Abstract

We study a random sequential adsorption (RSA) problem in which the packed objects are extended worldlines threaded through a closed conformally-expanding loop: per spatial axis, $x(z) = a\sin(bz) + a_2\sin(z)$ with integer frequency b and a unit slope budget (a "speed of light"), packed to a jamming cutoff under a mutual exclusion rule enforced at every timestep. The jammed count grows as a clean power law $N_{\text{sat}} \sim T^D$ in the time resolution T, with D well below the spatial dimension d: on the periodic (torus) form of the model we measure $D = 2.33$ in $d = 3$ and 1.38–1.42 in $d = 2$. We show that this exponent is a genuine *packing-number* exponent rather than the geometric dimension of any static configuration: the equal-time slices of the jam are *exactly* space-filling and homogeneous (wrapped correlation dimension $D_2 = 3.00$ in $d = 3$ and 2.00 in $d = 2$; rms comoving spread flat at $\sqrt{1/3}$ over the entire loop), and box-counting the full worldline bundle sweeps from 1 to $d{+}1$ with no plateau at D. An earlier apparent fractal dimension $D_2 \approx 2.79$ in the hard-wall form of the model is traced quantitatively to a boundary artifact of the correlation estimator. Finally, we extend the model with a free phase on even-frequency components — a deformation that specifically removes the kinematic reason the packing prefers odd frequencies — and find the packed state *structurally blind* to it: the exponents, slice geometry, and even/odd composition of the jam are unchanged, with only the jam density rising (+2.8% in $d = 2$, +6.9% in $d = 3$). The nontrivial content of the model appears to reside entirely in the whole-loop exclusion constraint.

1 Introduction

Random sequential adsorption — irreversible random placement of mutually excluding objects, run to jamming — is a classical subject with exact results in one dimension [1] and a large literature on jamming densities and kinetics for compact objects such as disks, spheres, and aligned squares [2, 3]. The objects packed here are not compact: they are *worldlines*, one-dimensional trajectories threaded through a closed "expanding then contracting" loop, which must avoid every other accepted worldline at *every* time slice simultaneously. Each trajectory is a two-term sinusoid per spatial axis, with an integer frequency and a slope constraint playing the role of a speed limit.

The construction is motivated by a toy cosmology — the loop coordinate z is conformal time on a closed universe with scale factor $a(z) = \sin z$, and the packed worldlines are "matter"

— but the questions we ask of it are combinatorial and geometric, and we present it here as a packing problem:

1. How does the jammed count N_{sat} grow with the time resolution T (equivalently, with the inverse exclusion size)? Is the growth exponent D equal to the spatial dimension d, or sub-dimensional?

2. Is D the fractal dimension of anything — a slice, or the worldline bundle — or is it a packing-number exponent with no geometric carrier?

3. Which microscopic degrees of freedom does the jam actually care about? In particular, the integer frequencies carry a parity structure (odd frequencies are at rest at the loop's turnaround, even ones are not), and one can ask whether the packing selects on it for kinematic or for global-geometric reasons.

Our answers, in brief: the growth is a clean power law with D substantially below d (Section 4); D is *not* a geometric dimension — on the periodic form of the model the equal-time slices are exactly space-filling and statistically identical at every conformal time, and the bundle's box-counting dimension has no plateau at D (Sections 6–7); and the parity structure of the jam turns out to be completely insensitive to a deformation (free phases) that removes its kinematic explanation, forcing a whole-loop-geometric one (Section 8.1). Along the way we document a boundary artifact of the standard correlation-dimension estimator that produced a convincing—and wrong—fractal dimension 2.79 on early versions of the model (Section 5); we include it because the diagnosis (probe-shape splitting of D_2 as an edge effect) seems useful beyond this model.

All data, code (CUDA engines, orchestration, analysis), and a dated lab notebook from which this account is assembled are public.[1]

2 The model

2.1 Worldlines

Time is a coordinate $z \in (0, \pi)$; the scale factor is $a(z) = \sin z$, so the loop expands from a "bang" at $z = 0$ to a turnaround at $z = \pi/2$ and recontracts to a "crunch" at $z = \pi$. A worldline has one component per spatial axis,

$$x(z) \;=\; a\,\sin(bz) \;+\; a_2 \sin z, \qquad b \in \mathbb{Z}_{\geq 2}, \tag{1}$$

with independent parameters per axis ($d = 2$ or 3 axes; we call these the 2+1 and 3+1 models). Both terms vanish at $z = 0, \pi$: every worldline meets the bang and the crunch. Positions are compared in *comoving* coordinates $X = x/\sin z$, in which the a_2 term is simply a constant offset,

$$X(z) \;=\; a\,\frac{\sin(bz)}{\sin z} \;+\; a_2. \tag{2}$$

Two forms of the amplitude constraint are studied:

[1] https://github.com/universe-analysis/universe-generator; the lab notes are published under docs/lab-notes/.

- **Hard-wall (original) model.** A joint slope budget $|ab| + |a_2| = 1$ per axis — the "speed of light" constraint — which also confines $|X| \leq 1$: comoving space is a box with hard walls. The budget couples position to speed: a worldline sitting far from the origin (large $|a_2|$) is necessarily slow (small $|ab|$).
- **Torus model.** The budget binds the wiggle term alone, $|ab| = 1$, and a_2 is drawn uniformly on $(-1, 1)$ as a free comoving offset. In comoving coordinates a uniform a_2 is exactly the homogeneous measure. Since a full-budget wiggle plus a free offset excurses past $|X| = 1$, the domain is closed into a period-2 torus: positions wrap, exclusion uses the minimum image, and there is no wall and no preferred center — these structures become *unstatable* rather than being removed by hand.

2.2 Exclusion and jamming

Sample z at T evenly spaced timesteps. Two worldlines *collide* if at *any* timestep their comoving positions fall within CELL $= 2/T$ of each other in the Chebyshev (L^∞) metric, per axis, minimum-image on the torus. Candidates are drawn at random and accepted iff they collide with no previously accepted worldline — textbook RSA, except that the excluded region of a candidate is determined jointly by *all* T timesteps.

True jamming is logarithmically unreachable at large T, so runs stop at an acceptance-rate cutoff (fraction of recent candidates accepted): 10^{-7} for the original campaigns, 10^{-6} for the torus-era campaigns after a dedicated study showed a cutoff decade moves the measured exponent by less than 0.02 in $d = 3$ (about 0.04 in the $d = 2$ torus model) while the count moves only through a T-independent prefactor ($N_{\text{sat}}^{10^{-6}}/N_{\text{sat}}^{10^{-7}} \approx 0.82$ flat across T) [2].

Frequencies are drawn from the full Nyquist band $b \in [2, T]$ with a high-frequency bias and a pairwise-coprimality constraint among a worldline's axis frequencies ("smart" sampling; it accelerates packing ~40% and does not change the measured exponents).

2.3 Phase degrees of freedom

The third model variant adds a free phase to the wiggle term,

$$x(z) \;=\; a\,[\sin(bz + f) - \sin f] \;+\; a_2 \sin z, \qquad f \sim U[0, \pi) \text{ iff } b \text{ even}, \tag{3}$$

the $-\sin f$ offset re-pinning the component to zero at the endpoints. Odd frequencies must remain phase-free for the worldline to close at $z = \pi$ ($\sin(b\pi + f) = \pm \sin f$ only cancels the offset for even b). The significance of the deformation is kinematic: without phases, a component's turnaround velocity is $\dot{X}(\pi/2) \propto ab\cos(b\pi/2)$, which vanishes for odd b and is maximal for even b — even-frequency worldlines are forced movers. With a phase, an even component's turnaround speed acquires a factor $|\cos f|$, so a phase near $\pi/2$ lets an even worldline sit arbitrarily close to rest. If the packing's observed preference for odd frequencies (Section 8) were a turnaround-kinematics effect, phases should dissolve it.

3 Numerical method

Packing runs on CUDA engines (one for $d = 2$, one for $d = 3$). The collision test uses a per-timestep uniform spatial hash in comoving coordinates: because the exclusion threshold

[2] A constant multiplicative shift in N_{sat} moves the intercept of $\log N_{\text{sat}}$ vs $\log T$, not the slope.

in *physical* coordinates is $\sin(z) \cdot 2/T$, it collapses in comoving coordinates to the constant $\mathrm{CELL} = 2/T$, giving exactly T cells per axis per timestep and an $O(NT)$ (rather than $O(N^2T)$) test. On the torus the hash and the pair test wrap mod T (minimum image). Campaigns are dispatched over a small GPU fleet by a resumable orchestrator; each accepted worldline's parameters are dumped so that any configuration can be reconstructed analytically at any z offline. $d = 3$ memory scales as T^4 (the dense per-timestep grid), capping consumer GPUs near $T \approx 200$; $d = 2$ scales as T^3 and reaches $T = 400$ cheaply.

The campaigns quoted below are: hard-wall $d = 3$, $T = 20$–200 step 10, 5 seeds per T, at both cutoffs; hard-wall $d = 2$, $T = 40$–400 step 20, 8 seeds; and the same grids for the torus model and the torus+phase model (the latter at cutoff 10^{-6} with a matched-cutoff no-phase baseline). Exponents are fit as weighted log–log slopes with seed bootstrap errors; quoted uncertainties are statistical only.

Estimators. (i) The *jam-count exponent*: the slope D of $\log N_{\mathrm{sat}}$ vs $\log T$, with power-law quality checked via the flatness of local (pairwise) slopes. (ii) The *correlation dimension* D_2 of an equal-time slice, from the Grassberger–Procaccia integral [4] $C(r) \sim r^{D_2}$ over a fixed comoving window, with both spherical (L^2) and cubic (L^∞) probes, in two variants: the standard open-boundary estimator, and a wrapped (minimum-image, periodic) estimator appropriate to the torus [3]. (iii) *Box-counting* dimensions of slices and of the full worldline bundle (in d and $d+1$ dimensions), reported as local slope vs scale — box-counting converges far too slowly in T to be used as a headline number, and we use it diagnostically. (iv) *Generalized dimensions* D_q from the q-th order correlation integrals [5].

4 The jam-count exponent

The count $N_{\mathrm{sat}}(T)$ is a clean power law in every variant measured (Fig. 1): local slopes are flat across the ladder with no crossover, fits are cutoff-robust in the sense quantified above, and the result is insensitive to the shape of the exclusion region (replacing the Chebyshev cube by an L^2 ball moves converged dimensions by < 0.005; Fig. 2). The exponents:

	$d = 2$	$d = 3$
hard-wall	1.58	2.46
torus	1.42 (10^{-7}) / 1.38 (10^{-6})	2.28 full / 2.33 ($T \geq 100$)
torus + phases	1.40 / 1.38 ($T \geq 100$)	2.27 full / 2.33 ($T \geq 100$)

Statistical errors are ± 0.002–0.006. Three observations:

1. D is well below d in every case: pure spatial packing would give $N_{\mathrm{sat}} \sim T^d$ (each worldline excludes $O(1)$ cells of a T^d grid), and the deficit is the cost of the joint across-time exclusion.

2. Moving from the hard-wall to the torus model lowers D by ≈ 0.15 in both dimensions (-0.16 in $d = 2$, -0.14 in $d = 3$ at high T). The mechanism reads off the constraint change: under the joint budget $|ab| + |a_2| = 1$ a worldline could buy a small wiggle by sitting far out (a "cheap wall-hugger" with tiny footprint), and such worldlines padded the

[3] Implemented with a periodic k-d tree; on a uniform periodic control cloud the wrapped estimator returns the exact ambient dimension for both probe shapes.

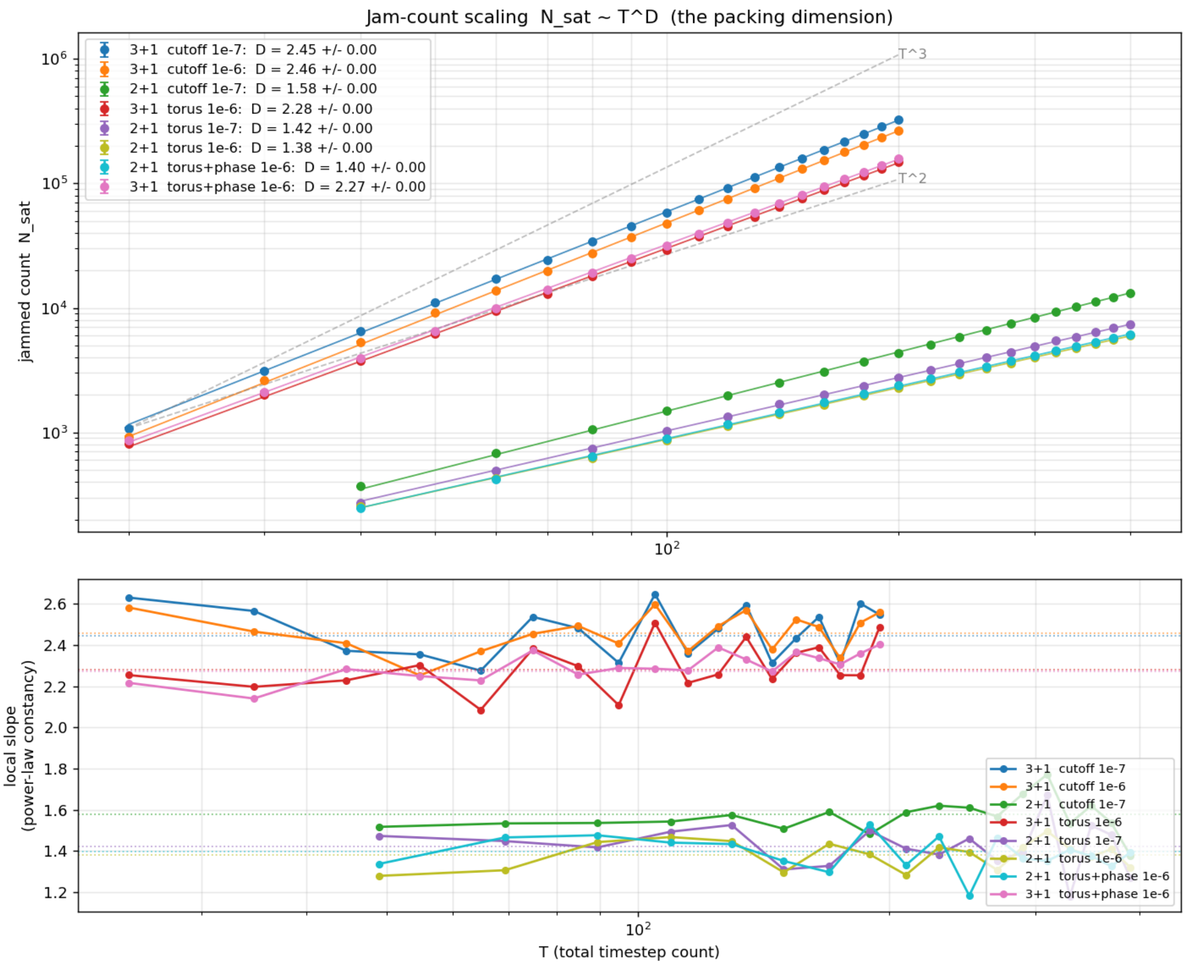

Figure 1: Jam count N_{sat} vs resolution T (log–log) for all model variants, with T^2 and T^3 guides. Top group: $d = 3$ (hard-wall at two cutoffs, torus, torus+phase); bottom group: $d = 2$. Lower panel: local slopes — flat, i.e. clean power laws, for every dataset. The torus and torus+phase ladders lie on top of each other at a slope visibly below their hard-wall counterparts.

count. On the torus every worldline carries the full unit swing, so the jam admits fewer objects and grows more slowly. The similarity of the shift across d suggests it is a property of the budget, not of the embedding dimension.

3. Adding phases does not move the exponent in either dimension (the differences are at the 0.01 level with mixed sign), and in $d = 3$ the phased ladder is the *cleanest* power law measured (local-slope scatter 0.066 vs 0.114 without phases), landing exactly on the 2.33 that the no-phase high-T fit suggested as the converged value. What phases do change is the prefactor: at fixed T the phased jam admits more worldlines (+2.8% at $T = 400$ in $d = 2$; +6.9% at $T = 200$ in $d = 3$). Phases are a packing-density knob, not a scaling knob.

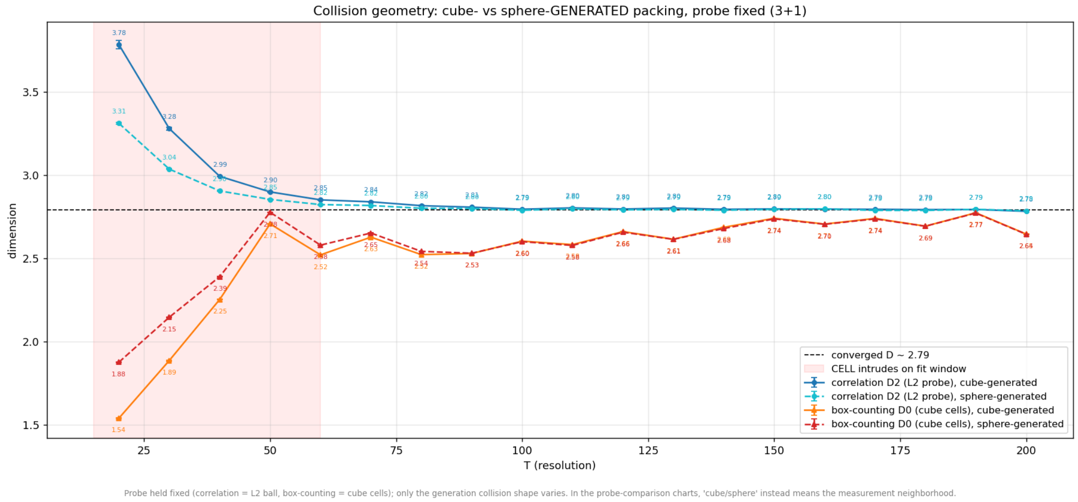

Figure 2: Exclusion-shape control ($d = 3$, hard-wall): correlation D_2 and box-counting D_0 measured on cube-collision (solid) vs sphere-collision (dashed) packings, probes held fixed. Converged over $T \geq 100$ the collision shape moves D_2 by 0.004 and D_0 by 0.003 — the dimension does not care about the shape of the excluded region.

5 An instructive artifact: the "fractal" hard-wall slice

The natural static object to interrogate is the turnaround slice: each worldline's comoving position at $z = \pi/2$, i.e. the matter distribution at maximum expansion. On the hard-wall model, the standard correlation estimator gave a stable, seed-robust, resolution-independent $D_2 = 2.79$ (spherical probe) — convincingly fractal-looking, and for some time the headline number of the project. Two anomalies eventually broke it: the cubic probe read 2.73 (a probe-shape split that a true dimension should not have), and the q-order spectrum D_q showed only mild multifractality ($D_1 \approx 2.80$ down to $D_5 \approx 2.72$), too weak to explain the split.

The resolution is that the standard estimator $C(r) = \Pr(\text{pair within } r)$ silently assumes an unbounded or edge-corrected sample. The hard-wall model is neither: $|X| \leq 1$ is a hard boundary and the sampler piles density against it, so every near-wall point is missing the neighbours that would lie outside, suppressing $C(r)$ at large r and dragging the fitted slope down. Controls make this quantitative: on a *uniform* cloud (true $D_2 = 3$) the naive estimator returns 2.87 (sphere) / 2.82 (cube) — wrong, and split by the same $\sim$0.05 — while interior-only centers or periodic wrapping restore 3.000 for both probes with the gap collapsed. The probe-shape split is thus itself an edge diagnostic: corners of the L^∞ ball reach $\sqrt{d}\,r$ into the depleted zone, so the cube probe is hurt more, and indeed the split shrinks from $\sim$0.06 in $d = 3$ to $\sim$0.025 in $d = 2$ exactly as corner reach predicts.

We flag this because the artifact was stable under every robustness test short of the boundary controls — seeds, resolution, cutoff, collision shape — and produced a plausible fractal dimension out of thin air. On a bounded, edge-crowded sample, an uncorrected pair-correlation dimension should be treated as a lower bound at best.

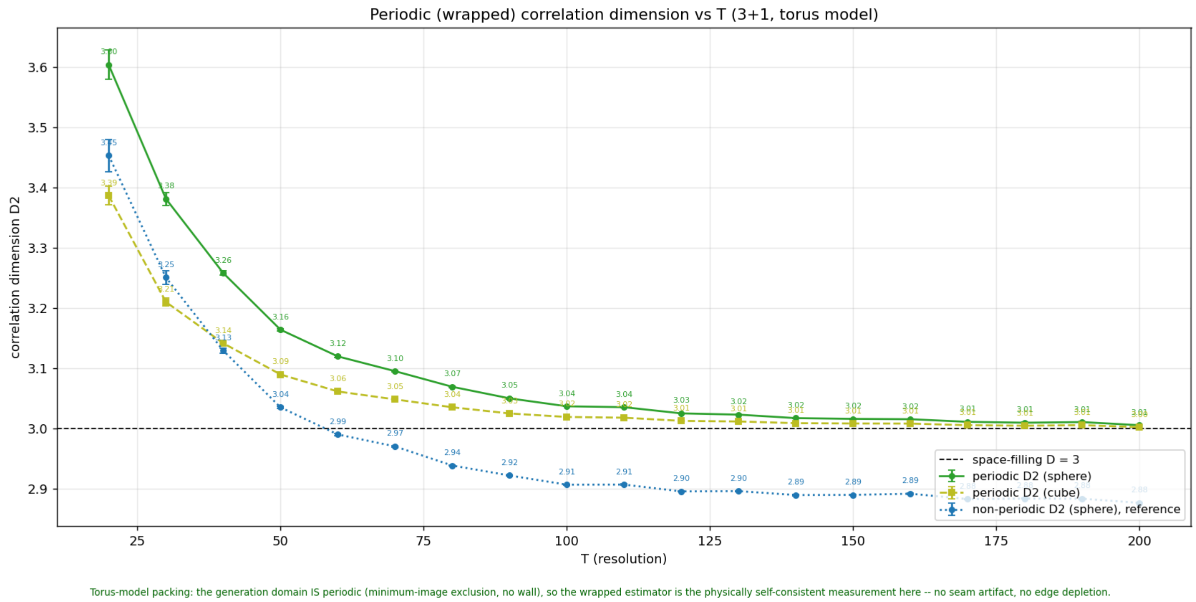

Figure 3: Wrapped (minimum-image) correlation dimension of the turnaround slice vs T for the $d = 3$ torus+phase model (5 seeds per T): spherical and cubic probes converge together onto the space-filling $D = 3$ line (3.01 at $T = 200$), while the open-boundary estimator on the same clouds reads ∼2.89 — the boundary artifact of Section 5, reproduced on demand. The no-phase torus dataset is indistinguishable point by point.

6 The torus jam is exactly homogeneous

On the torus model the domain is genuinely periodic, the wrapped estimator is the physically self-consistent one, and the verdict is unambiguous (Fig. 3): the turnaround slice is *space-filling*,

$$D_2^{\text{wrapped}} = \begin{cases} 3.019 \text{ (sphere)}/3.010 \text{ (cube)}, & d = 3 \text{ (with phases: } 3.021/3.011 \text{ without)}, \\ 2.032/2.025, & d = 2 \ (2.033/2.026 \text{ without}), \end{cases}$$

seed-averaged over $T \geq 100$, descending onto $D = d$ with the probe-shape gap collapsed. The homogeneity is not special to the turnaround: fixing one universe ($T = 200$) and scanning all conformal times, the rms comoving spread sits at $\sqrt{1/3} \approx 0.5774$ — the exact value for a uniform torus — flat across the entire loop through bang, turnaround, and crunch, and the slice box-counting dimension stays within a 2.61–2.66 band (Fig. 4; the residual deficit from 3 is the usual slow convergence of box-counting, not structure). The uniform-offset proposal measure makes homogeneity true of the *candidate* distribution by construction; the measurement shows that jamming *preserves* it — exclusion thins the cloud without structuring it at these scales.

Together with Section 4 this cleans up the model's story: the geometry of any single slice is featureless (no fractality, no center, no wall), and *all* of the nontrivial structure lives in the packing-number exponent. The earlier 2.79 was the hard-wall model's boundary structure talking, not the packing.

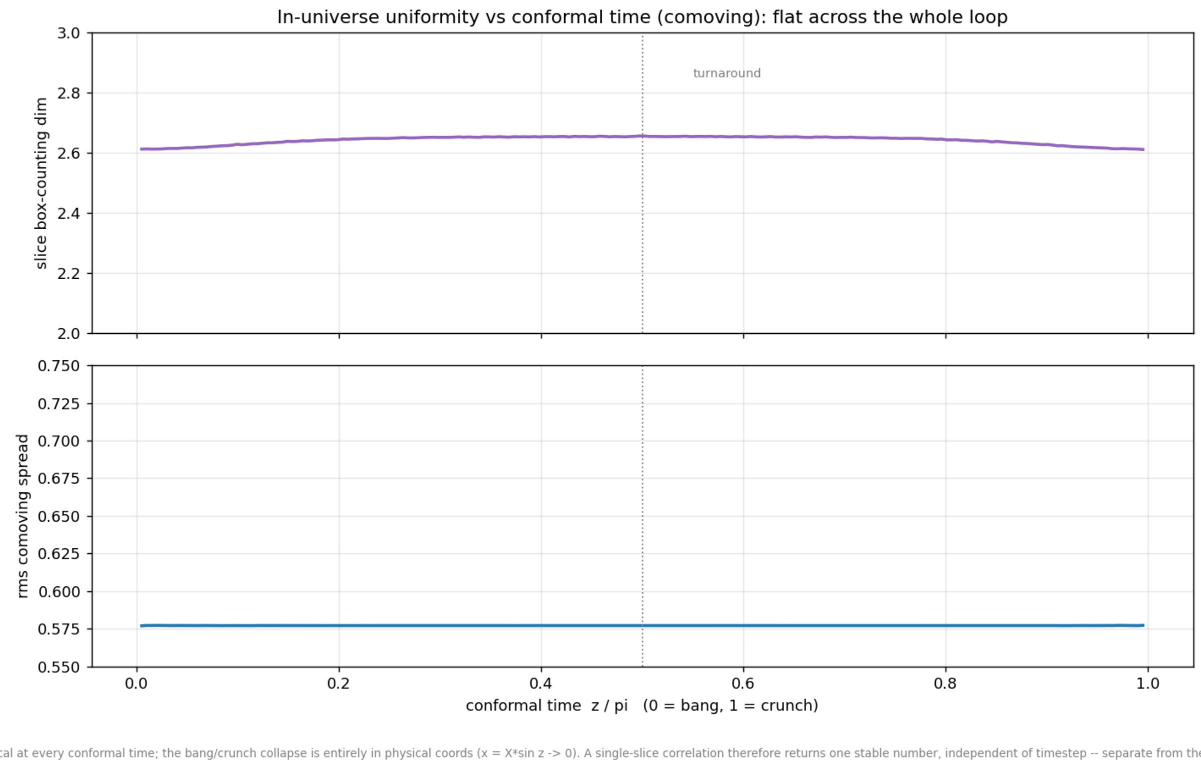

Figure 4: In-universe uniformity over conformal time ($d = 3$ torus with phases, $T = 200$, 5 seeds): slice box-counting dimension (top) and rms comoving spread (bottom) vs z/π. The spread sits on $\sqrt{1/3}$ (uniform torus value) over the whole loop.

7 D is not the dimension of anything

Since a slice is exactly d-dimensional, one might still hope the count exponent is the fractal dimension of the full worldline *bundle*. It is not. Box-counting the pooled bundle (all timesteps, $d = 3$) gives a local dimension that runs from ≈ 1 at fine scales — where individual smooth curves are resolved — to ≈ 3 at coarse scales, crossing 2.33 only in passing with no plateau (Fig. 5). The honest $(d+1)$-dimensional version (conformal time as a fourth axis, under both a native and a cell-isotropic time scaling) does the same, sweeping $1 \to 4$ through 2.33 without pausing.

For *point* packings, packing number and box dimension coincide, which is why the intuition expects agreement. For packings of extended curves under a joint across-time constraint they need not, and here they demonstrably do not: the spatial arrangement is uniform (geometric dimension $\to d$), so the sub-T^d count is not spatial clustering — it is the across-time exclusion suppressing how many worldlines can be threaded through *all* timesteps at once. D is well-defined as a count exponent, with no static geometric carrier. We are not aware of a standard name for this object in the RSA literature; "packing-number exponent of a curve packing" seems accurate.

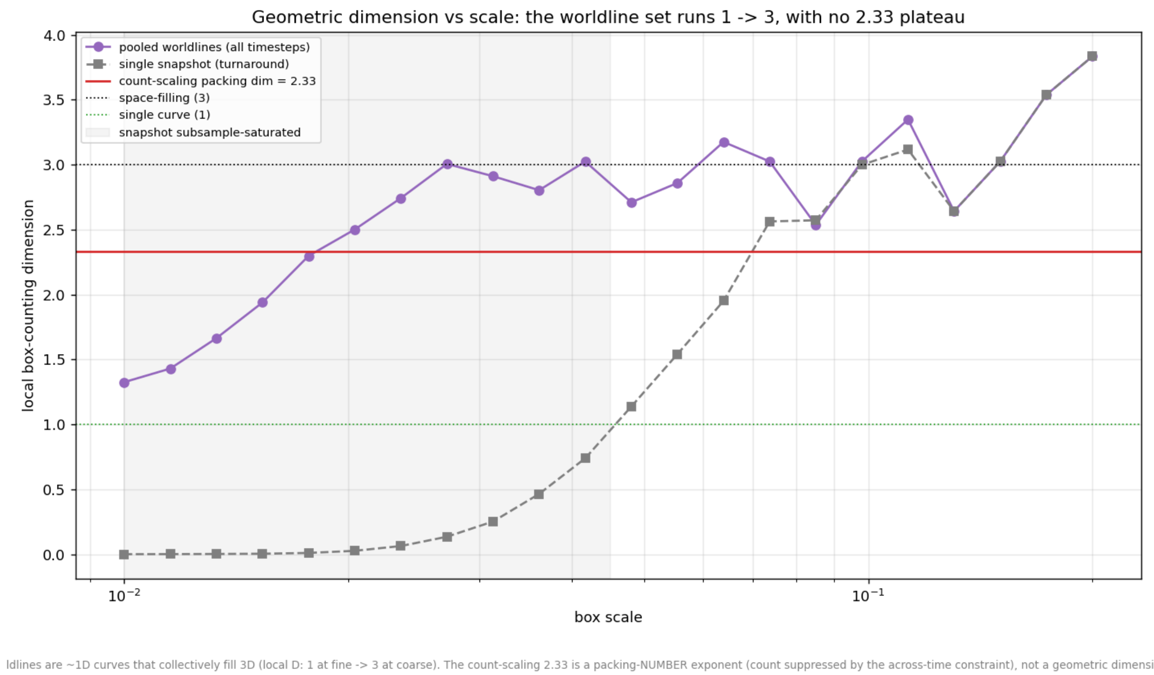

Figure 5: Local box-counting dimension vs scale for the pooled worldline bundle ($d = 3$ torus, $T = 200$, all timesteps pooled into one $\sim 10^7$-point cloud, seed-averaged). The local dimension sweeps from ≈ 1 (resolving individual smooth curves) to ≈ 3 (space-filling), crossing the count exponent 2.33 (line) only in passing, with no plateau.

8 Frequency parity: structure the jam does have

The comoving wiggle $\sin(bz)/\sin z$ is symmetric about the turnaround for odd b (the worldline sits at a velocity extremum — at rest) and antisymmetric for even b (it swings through at speed). The jam has a definite composition along this axis: at $T = 200$ ($d = 3$, torus), 47–48% of packed worldlines are entirely odd-frequency, and 82–83% of all accepted frequencies are odd, both essentially flat in T above ~ 50 (Fig. 6, left and right). Acceptance is also stage-resolved: in the sparse early quartile of the acceptance sequence, even frequencies are accepted at 20–27% across the whole band, while in the final near-jam quartile they are squeezed to 3–13%, recovering with b (Fig. 7) — odd worldlines are what still fits into the last gaps.

In the hard-wall model this parity structure had a sharp kinematic signature: an even component's swing amplitude scales as $(1 - |a_2|)/b$, so only high-frequency evens could thread the packing at all, and below $T \approx 40$ (where the Nyquist band contains no sufficiently large even integer) the turnaround population was *exactly* 100% at rest — a geometric freeze-out with a near-discontinuous onset at $T \approx 40$–48, after which the moving fraction grew toward ~ 40–50%. The "forbidden zone" (no evens below $b \approx 34$) is a wall-era feature: on the torus, sparse-stage evens appear at low b freely (Fig. 7), though near-jam suppression of low evens survives.

8.1 Phases: a targeted null result

The phase deformation of Section 2.3 was designed against exactly this structure: it hands every even-frequency component a free phase, letting it sit as close to rest at the turnaround

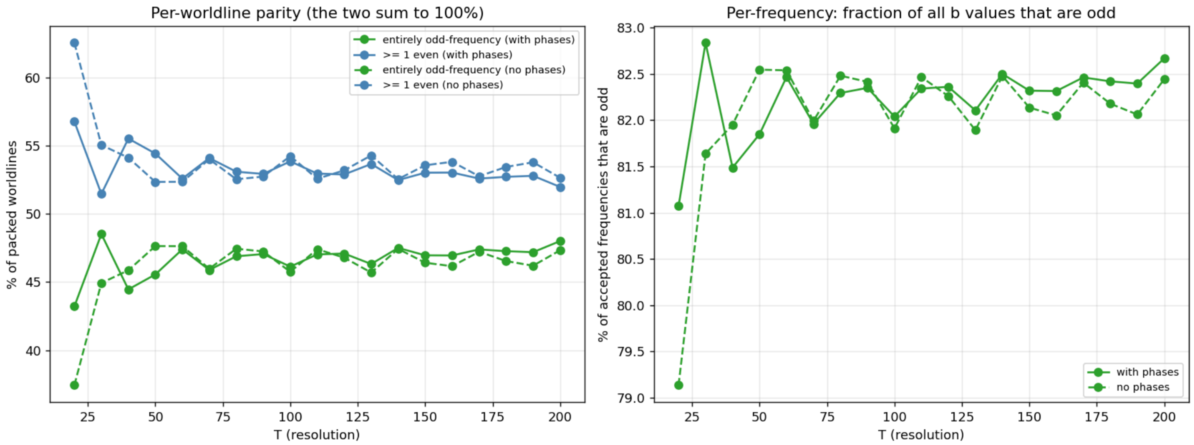

Figure 6: Parity composition of the jam vs T ($d = 3$ torus), phased (solid) vs no-phase (dashed). Left: fraction of worldlines entirely odd-frequency vs carrying at least one even frequency. Right: fraction of all accepted frequencies that are odd. The phased and unphased curves coincide at every T.

as it likes, and thereby removes the kinematic penalty evens pay. If the odd preference were turnaround kinematics, phases should dissolve it; if the jam merely needed *slow-at-turnaround* objects, phased evens are now such objects.

The measurement is a null across the board. With phases on (all at matched cutoff, same grids and seeds): the count exponents are unchanged (Section 4); the wrapped slice dimension is unchanged (3.019 vs 3.021, 2.032 vs 2.033); the uniformity profile over the loop is unchanged; and — most pointedly — the parity composition is unchanged at every T and at every packing stage (Figs. 6–7): 48.0% vs 47.3% all-odd worldlines, 82.7% vs 82.4% odd frequencies, and coincident stage-resolved acceptance curves. The only effect of the added degrees of freedom is the jam density prefactor (+2.8% / +6.9%).

The conclusion is that the odd-frequency preference is *not* a turnaround-kinematics effect. What parity controls that a phase cannot undo is the *whole-loop* shape of the trajectory — how the swing budget and the coprimality constraint thread a component through every timestep simultaneously. This is consistent with everything else in the paper: each structural readout of the jam (exponent, slice geometry, composition) is set by the global exclusion constraint, and deformations that act locally in time change only how densely the same structure packs.

9 Physical readouts of the hard-wall packing

For completeness we record the physical observables extracted from the hard-wall configurations, since they motivated much of the above (all at cutoff 10^{-7}, $d = 3$; these predate the torus model and should be read as properties of the wall-era packing). Interpreting worldline energy through either of two dictionaries ($E \propto b$, "quantum wave", or $E \propto$ arc length, "string") gives dictionary-robust results:

- **Equation of state.** With $w = P/\rho = \frac{1}{3}\langle Ev^2\rangle/\langle E\rangle$ and the slope-1-capped velocities, the turnaround state is matter-like: $w = 0.042$ under both dictionaries (agreeing to 1.6%).

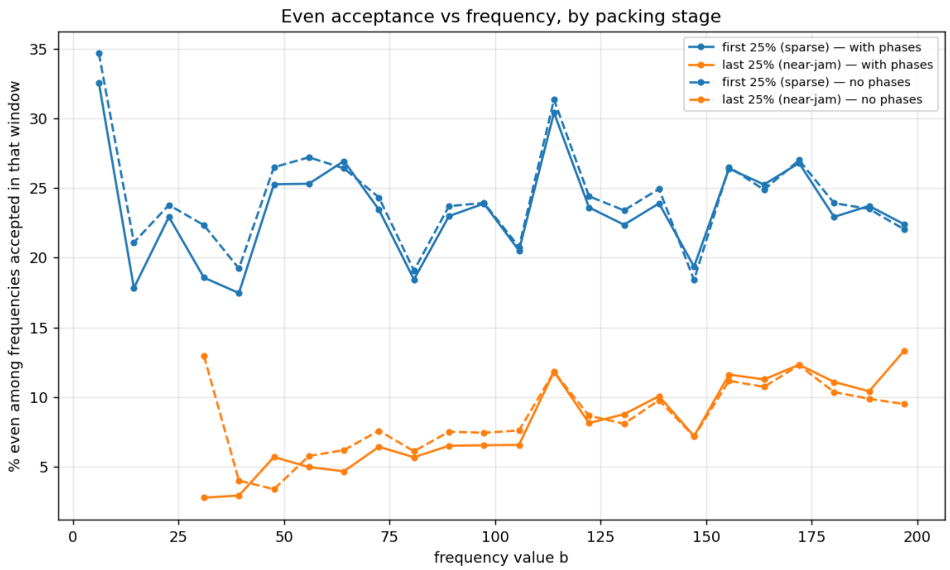

Figure 7: Even acceptance vs frequency value, resolved by packing stage ($d = 3$ torus, $T = 200$, full ordered acceptance sequences, $N \approx 1.5 \times 10^5$): first quartile (sparse, blue) vs last quartile (near-jam, orange), phased (solid) vs no-phase (dashed). Near jamming the survivors skew hard to odd at every b; phases change neither curve.

- **History.** Evaluating the same w at every conformal time gives a symmetric "bathtub": stiff ($w \approx 0.5$–0.65) near the bang and crunch, crossing radiation ($w = 1/3$) on the way in and out, with a razor-thin matter-like cusp ($w \approx 0.04$) exactly at the turnaround — a coherent closed-universe history, and dictionary-robust across the entire cycle.
- **Speed distribution.** The moving component's turnaround speeds peak low ($\sim$0.1) and terminate in a hard cutoff exactly at 1.0: the slope budget acts as a clean relativistic cap.
- **Knot selection.** Using the reduced torus-knot size $K(p, q) = (p/g - 1)(q/g - 1)$, $g = \gcd(p, q)$, as a knottiness proxy for axis-frequency pairs: coprimality statistics alone are scale-invariant (coprime fraction $6/\pi^2$, $\langle K \rangle \propto N^{2.01}$ for random pairs — no number-theoretic growth exponent), but the *packed* population runs 28–35% knottier than its proposal distribution, with the enrichment growing with T before saturating. Jamming selects for high-frequency, coprime, knottier worldlines — the frequency-drives-packing story seen through a topological proxy.

10 Discussion and open problems

The picture that survives all measurements to date: a jam of sinusoidal worldlines on the periodic loop is *geometrically featureless* — exactly homogeneous slices at every conformal time, space-filling correlation dimension, no fractal carrier for the count exponent — while the count itself grows sub-dimensionally with a robust, variant-stable exponent (2.33 in 3+1, 1.38–1.42 in 2+1 on the torus). The exponent quantifies the dynamical cost of threading extended objects through every time slice at once, and it is remarkably indifferent to microscopic deformations: exclusion shape (cube vs sphere), stopping cutoff, and rest-enabling phases all leave it fixed, the last of these also leaving the jam's parity composition untouched at every packing stage.

Open problems we consider sharp:

1. **Is there an exact value?** $D = 2.33 \approx 7/3$ in 3+1 and 1.38–1.42 in 2+1 at the accessible cutoffs; whether these converge to rational values (and what fixes them) is open. An analytic attack on a simplified version (e.g. frequency-1 worldlines only, or a 1+1 reduction) may be tractable.

2. **The cutoff-to-jamming extrapolation.** The exponent is cutoff-stable to < 0.02/decade, but a principled extrapolation to true jamming (Feder-law style [2]) has not been done for curve packings.

3. **Why is parity composition invariant under phases?** The null is clean but unexplained: the whole-loop constraint preserves the even/odd acceptance statistics *exactly* while admitting 7% more objects. A mean-field model of the across-time exclusion that predicts both the exponent shift under the budget change (-0.15) and the parity invariance would be substantial progress.

4. **Kinetics.** The attempts-to-jam cost grows as roughly T^{10} — far steeper than N_{sat} — and the approach-to-jamming exponent for extended worldlines (the analog of the Feder law) is unmeasured.

Reproducibility. Every figure in this paper is committed to the project repository alongside the code and the exact commit that generated it (the lab-notes convention); the packing engines, the orchestration layer, and all estimators are public at the repository above.

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